Fixed Wing UAV



Category: **Drones & Aviation** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Fixed Wing UAV

What it is used for

A fixed-wing UAV is an unmanned aircraft that uses wings to generate lift for flight. Operate it only with proper training, legal authorisation, approved airspace, and a mission-specific flight plan.

Main parts

- Fixed wings and fuselage
- Propulsion system, motor, and propeller
- Flight controller and navigation sensors
- Battery or approved power system
- Radio transmitter, receiver, and antennas
- Landing gear and payload mount, if fitted

Before you start

1. Confirm required aviation permissions, airspace restrictions, and local operating rules.
2. Inspect wings, fuselage, control surfaces, fasteners, and landing gear.
3. Check the propeller, motor mount, and propulsion system for damage.
4. Inspect the battery, connectors, and wiring for damage or swelling.
5. Verify control links, navigation sensors, failsafe settings, and battery status.
6. Review weather, wind, visibility, flight route, landing area, and emergency plan.

How to use

1. Select a clear approved operating area away from people and obstacles.
2. Install the battery and payload according to the manufacturer's instructions.
3. Secure all covers, fasteners, and connectors before powering on.
4. Place the UAV in the specified preflight position and power it on.
5. Complete controller checks, sensor checks, and navigation readiness checks.
6. Verify control-surface direction and motor operation using the approved procedure.
7. Confirm home position, failsafe behaviour, return procedure, and battery reserve.
8. Start the flight only when the area is clear and all checks pass.
9. Launch using the approved method for that aircraft and conditions.
10. Monitor flight path, battery, telemetry, weather, and surrounding airspace continuously.
11. Abort or return to the landing area if conditions become unsafe.
12. Land using the approved procedure and disarm the propulsion system.
13. Power off the aircraft and controller as instructed, then remove the battery if required.

4. Inspect the UAV after flight and record any faults or unusual behaviour.

⚠ Safety

- Only trained and authorised operators should fly the UAV.
- Follow applicable aviation rules, airspace restrictions, and required permissions.
- Keep people clear of the propeller and never handle it while powered.
- Do not fly in unsuitable weather, poor visibility, or unsafe wind conditions.
- Maintain required separation from people, buildings, aircraft, and restricted areas.
- Do not continue flight with low battery, lost control link, or critical warnings.

Care & storage

- Clean dirt and moisture from the airframe after use.
- Store batteries in a safe, dry location according to battery instructions.
- Inspect propellers, control surfaces, fasteners, and connectors regularly.
- Update firmware only using approved procedures and verify settings afterward.
- Record flights, maintenance, battery condition, and faults.

Common mistakes to avoid

- Skipping preflight checks or failing to verify control-surface direction.
- Launching without checking airspace, weather, or landing conditions.
- Ignoring battery reserve, telemetry warnings, or failsafe alerts.
- Touching or adjusting the propeller while the propulsion system is armed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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